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NLP Library

What is spaCy?

- SpaCy is a free, open-source library for advanced Natural Language Processing (NLP) in Python.
- Fastest NLP Library in the world
- Easy to install with simple API
- Includes deep learning kits and incorporates TensorFlow, Keras, Scikit-Learn, Gensim, etc. flawlessly.

Features of spaCy

NAME	DESCRIPTION
Tokenization	Segmenting text into words, punctuations marks etc.
Part-of-speech (POS) **Tagging**	Assigning word types to tokens, like verb or noun.
Dependency Parsing	Assigning syntactic dependency labels, describing the relations between individual tokens, like subject or object.
Lemmatization	Assigning the base forms of words. For example, the lemma of "was" is "be", and the lemma of "rats" is "rat".
Sentence Boundary Detection (SBD)	Finding and segmenting individual sentences.
Named Entity Recognition (NER)	Labelling named "real-world" objects, like persons, companies or locations.
Entity Linking (EL)	Disambiguating textual entities to unique identifiers in a knowledge base.
Similarity	Comparing words, text spans and documents and how similar they are to each other.
Text Classification	Assigning categories or labels to a whole document, or parts of a document.
Rule-based Matching	Finding sequences of tokens based on their texts and linguistic annotations, similar to regular expressions.
Training	Updating and improving a statistical model's predictions.
Serialization	Saving objects to files or byte strings.

Linguistic Annotations

SpaCy provides a variety of linguistic annotations to give you insights into a text's grammatical structure. This includes the word types, like the parts of speech, and how the words are related to each other.

Stemming

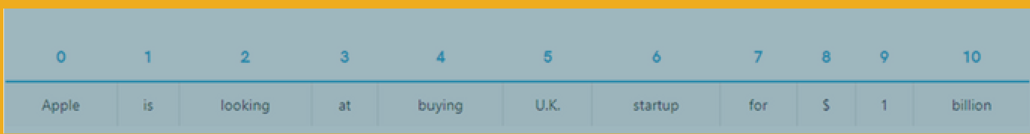
Stemming refers to reducing a word to its root form. The stem does not have to be a valid word at all. Stemming algorithms remove affixes (suffixes and prefixes). For example, the stem of "university "is "univers".

Lemmatization

A lemma is the base form of a token. The lemma of walking, walks, walked is walk. Lemmatization is the process of reducing the words to their base form or lemmas.

Tokenization

During processing, spaCy first tokenizes the text, i.e. segments it into words, punctuation and so on. This is done by applying rules specific to each language. First, the raw text is split on whitespace characters then the tokenizer processes the text from left to right.



0	1	2	3	4	5	6	7	8	9	10
Apple	is	looking	at	buying	U.K.	startup	for	$	1	billion

On each substring, it performs two checks:

- Does the substring match a tokenizer exception rule? For example, "don't" does not contain whitespace, but should be split into two tokens, "do" and "n't", while "U.K." should always remain one token.
- Can a prefix, suffix or infix be split off? For example punctuation like commas, periods, hyphens or quotes.

Picture Source

- unsplash.com
- pexels.com
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